

Machine Learning–Driven Materials Design for Additive Manufacturing

Project Description and Clarification

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Note: Recall that this writing assignment says:

Length = 2+ pages text plus appendices as needed. Cover page, table of contents, glossary, references, pictures, tables, images, diagrams do not count for the 2+ pages text.

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I. Introduction

For this Capstone Project we will be working with the Washington State University School of Mechanical and Materials Engineering. We will be building off a previous team's capstone project, fixing bugs, and expanding the capabilities already implemented. The existing project is a desktop program which accepts microscopy images of various materials and then identifies different inclusions in the material. These inclusions are normally a secondary phase of the surrounding compound or a different compound altogether, the addition of which changes the characteristics of the material.

In addition to the bug fixing, we plan on incorporating a forward model that will allow a user to input a microscopy image of a material, and the program will output a preliminary assessment of that material's physical characteristics. Should time permit, we also hope to implement an inverse model that will allow a user to input the desired physical characteristics of a material, and the program will specify the type and composition of the inclusions that will result in said characteristics.

II. Background and Related Work

This work is located in the intersection of materials science, electron microscopy image analysis, and machine learning, where the aim is to make quantitative connections between materials and their properties, such as strength and hardness. Current trends in image analysis, especially within image analysis in materials science, show a shift towards automatic image segmentation tasks to eliminate human bias and make image analysis more effective. Conventional approaches to image analysis include the development of hand-crafted features and traditional computer vision techniques. Research papers such as (Russ, 2011) show that grain boundary images are usually analyzed using hand-crafted features. Recent studies have aimed to improve image analysis accuracy by using deep learning paradigms such as CNNs to analyze images and extract better results compared to traditional computer vision techniques (DeCost & Holm, 2015). The development of foundation models such as Segment Anything Model (SAM) marks a milestone in image analysis, enabling domain-agnostic image segmentation with minimal training (Kirillov et al., 2023). Similarly, within materials informatics, studies such as (Agrawal & Choudhary, 2016) have proposed forward models and inverse models. However, the effectiveness comes at the cost of computational efforts, large amounts of labeled data, or applicability limitations for non-technical users. The work leverages all these contributions to provide an interface for vision model usage along with the implementation of Low-Rank Adaptation (LoRA) for efficient fine-tuning of vision models for forward and inverse material property predictions without the need for fully training the large vision models from scratch (Hu et al., 2021). Thus, the project aims to place itself in an applied context as an extension of leading techniques in ML along with experimental materials science. The technical skills that need to be obtained for completing this project by the team are image segmentation techniques, deep vision models as well as techniques for fine-tuning such models using transfer learning with techniques like LoRA for reduced parameter adjustments for fine-tuning large vision models without having to train the large vision models from scratch, as well as statistical representation of microstructural features along with proper evaluation techniques for ML models for testing on microscopy images.

III. Project Overview

This project involves creating a reliable and accessible software program geared towards effectively evaluating microscopic images of engineering materials with a purpose of relating microscopic materials to their macroscopic properties. In materials science and mechanical engineering, determining how microscopic-level structures and particles shape macroscopic-level factors such as strength, hardness, and durability is a critical measure, especially for those materials utilized under conditions of extreme stress such as in pressure vessels, reactors, ballistics, and even highly sophisticated manufacturing equipment. This is because existing techniques in this line are mostly arbitrary and heavily based on interpretation.

The project improves upon the existing Python-based desktop application developed by prior capstone teams. The application enables users to input electron microscope images of materials and allows users to perform image segmentation on those materials by integrating various techniques, such as machine learning and traditional image segmentation techniques, including a machine learning approach like the Segment Anything Model (SAM) for image segmentation. Following segmentation, statistical attributes characterizing secondary phases within the material are calculated, such as size, density, and location.

It has only been tested on existing data, which already existed in the dataset, and its results have always been unfair or inconsistent, especially after the introduction of new data. There are instances where segmentation happens, while in other instances there is no reason for the inconsistency. Thus, the main challenge in this project is not, as may be perceived, to completely redesign the system, but to find areas of weakness and improve the application to ensure consistency with any new data supplied by various research groups or institutions.

The overall scientific goals of this project include understanding how material processing, microstructure, and material properties are related to one another. This includes analyzing how different factors, such as how much power is being used in the laser, the size of the grains in the powder, and different forms of additive manufacturing, affect different microstructural features that are being viewed under a microscopy image. Of note is that much of this project is focused on understanding additively manufactured materials, as these have a tendency to have differing microstructures, making them more difficult to review. Both densely packed small inclusions and sparsely packed large inclusions are referenced in relation to these investigations.

IV. Client and Stakeholder Identification and Preferences

Our primary client is Arezoo Zare, an Assistant Professor at the Washington State University School of Mechanical and Materials Engineering. By extension, the School of Mechanical and Materials Engineering, part of the Voiland College of Engineering and Architecture here at Washington State, is our secondary client. Other current stakeholders include but are not limited to materials research teams and additive manufacturing research teams. Should our program eventually become polished enough for widespread use beyond research and academia, production teams of various affiliations (government, private industry, etc.), and anybody else who could utilize a program that can identify material characteristics from a microscopy image (forward model), and/or specify the microscopic material inclusions necessary for certain material characteristics (inverse model) would also become a shareholder.

Our primary and secondary clients' needs are primarily research oriented, as are a majority of our stakeholders. Their primary focus is the research of various materials, especially those made with additive manufacturing. Materials made with additive manufacturing can have vastly

different characteristics depending on their microscopic structure, inclusions (or lack thereof), and the composition of said inclusions (secondary phase of the surrounding substance or a different substance altogether). As such, being able to identify a material's characteristics via microscopy images (forward model) would prove immensely useful. Being able to specify desired characteristics and then be given a corresponding microscopic structure (inverse model) would also be useful in a similar manner.

Should our program prove useful in a research and academia setting, expanding its capabilities to a production level would allow it to be used outside of academia. The government and private industry especially could see use in this program in the development of new products and services, as well as in researching and improving existing products.

V. Glossary

Additive manufacturing: Also known as 3d-printing, this is the process of making something by adding successive layers of material.

Inclusion: A “pocket” of material that is different from the surrounding material in some fashion. These can be made on purpose or on accident and normally bring change the characteristics of the material as a whole in some way.

Secondary Phase: A different state of a similar base material, normally identifiable by a different microscopic structure. An “inclusion” of a secondary phase often imparts a different physical characteristic (their inclusion can make a material stronger or weaker, more or less brittle, etc).

Define technical terms used in the document.

VI. References

- Russ, J. C. (2011). *The Image Processing Handbook*. CRC Press.
- DeCost, B. L., & Holm, E. A. (2015). A computer vision approach for automated analysis and classification of microstructural image data. *Computational Materials Science*, 110, 126–133.
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- Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.